

Material Safety Data Sheet

Glufosinate-Ammonium 15% W/V SL

1. IDENTIFICATION OF THE SUBSTANCE

Product name: Glufosinate-Ammonium 15 Pct. W/V SL

Company Identification:

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2. HAZARD IDENTIFICATION

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 [EU-GHS/CLP]

Acute toxicity, Inhalation - category 4

Skin irritation-category 2

Eye irritation-category 2A

Skin sensitization-category 1

Reproductive toxicity - category 1B

Specific target organ toxicity (repeated exposure) - category 2

2.2 Label elements

Labelling according Regulation (EC) No 1272/2008 [CLP]



Pictogram

Signal word

Danger

Hazard statement(s)

H315: Cause skin irritation.

H319: Cause serious eye irritation.

H360: May damage fertility or the unborn child.

H373: May cause damage to organs through prolonged or repeated exposure.

H317: May cause an allergic skin reaction.

H332: Harmful if inhaled.

Precautionary statement(s)

P201: Obtain special instructions before use.

P202: Do not handle until all safety precautions have been read and understood.

P281: Contains refrigerated gas; may cause cryogenic burns or injury.

P280: Wear protective gloves/protective clothing/eye protection/face protection.

P264: Wash face, hands and any exposed skin thoroughly after handling.

P260: Do not breathe dust/fume/gas/mist/vapours/spray.

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P261: In contact with water releases flammable gases.

P271: May cause fire or explosion; strong oxidiser.

P272: May intensify fire; oxidiser

Response statement(s)

P308+P313: IF exposed or concerned: Get medical advice/attention.

P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P337+P313: If eye irritation persists: Get medical advice/attention.

P302+P352: IF ON SKIN: Wash with plenty of soap and water.

P362: Take off contaminated clothing and wash before reuse.

P304+P340: IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.

P314: Get medical advice/attention if you feel unwell.

P312: Call a POISON CENTER or doctor/physician if you feel unwell.

P321: Specific treatment.

P332 + P313: If skin irritation occurs: Get medical advice/attention.

P363: Wash contaminated clothing before reuse.

Storage and Disposal statement(s)

P405: Store locked up.

P501: Dispose of contents/container to an approved waste disposal plant.

2.3 Other hazards

Hazards not otherwise classified(HNOC): Not applicable.

Other Information: May be harmful in contact with skin. Harmful to aquatic life with long lasting effects.

Unknown acute toxicity: 0% of the mixture consists of ingredient(s) of unknown toxicity.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical characterization: phosphinic acid herbicide

Ingredient Name	CAS Number	Concentration(w/v)
Glufosinate-ammonium	77182-82-2	15%
Other Ingredients	-	To 100%

4. FIRST AID MEASURES

General information

Eyes: Flush with plenty of water. Get medical attention if irritation persists.

Skin: Remove contaminated clothing. Wash skin immediately with plenty of soap and water. Get medical attention if irritation occurs.

Ingestion: Rinse mouth thoroughly with plenty of water. Do not induce vomiting. Get medical attention immediately.

Inhalation: Remove individual to fresh air. Get medical attention if breathing difficulty develops.

Advice to doctor

Special notes to a physician: Endotracheal intubation and gastric lavage should be

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performed as soon as possible, followed by charcoal and sodium sulfate administration. Respiratory, Cardiac and Central Nervous Systems should be monitored with particular regard to ECG, electrolyte balance (especially for potassium) and signs of increased intracranial pressure. In the event of a large exposure, dialysis and/or hemoperfusion should be conducted as soon as possible to eliminate the compound from the body. In the event of convulsions, administer Phenobarbital or diazepam. There is no specific antidote. Glufosinate- ammonium does not inhibit cholinesterase; thus atropine and 2-PAM are contraindicated. Recovery is normally spontaneous, usually within 48h.

5. FIRE-FIGHTING MEASURES

Extinguishing media

Water fog, fine water spray, foam, dry chemical, carbon dioxide.

Hazards from Combustion products

In a fire, irritant and toxic fumes containing oxides of carbon, nitrogen, phosphorus and sulphur,

and other toxic substances may be generated.

Precautions for fire fighters

Fire fighters should wear full protective gear, including self-contained breathing apparatus (AS/NZS 1715/1716). Keep unnecessary people away. If it can be done safely, remove intact containers from the fire. Otherwise, use water spray to cool them. Bund area with sand or earth to prevent contamination of drains or waterways. Dispose of fire control water or other extinguishing agent and spillage safely later.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures:

Isolate hazard area. Keep unauthorized people away. Avoid contact with spilled product or contaminated surfaces.

Environmental precautions: Do not apply directly to water, or to areas where surface water is present. Do not apply to intertidal areas below the mean high water mark. Do not contaminate water by cleaning of equipment or disposal of equipment washwaters or rinsate.

Methods and materials for containment and cleaning up:

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Collect and transfer the product into a properly labelled and tightly closed container. Clean contaminated floors and objects thoroughly, observing environmental regulations.

7. HANDLING AND STORAGE

Handling

Keep out of reach of children. Harmful if absorbed by skin contact or if swallowed. Will irritate the eyes and skin. Avoid contact with eyes and skin. If product on skin, immediately wash area with soap and water. If product in eyes, wash it out immediately with water. Wash hands after use. After each day's use, wash gloves, face shield or goggles, respirator if worn, and contaminated clothing.

Storage

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Store in the closed, original container in a cool, dry, well-ventilated area. Do not store for prolonged periods in direct sunlight. Keep away from all ignition sources. Keep the container tightly closed, and protect from moisture. Do not store in steel or aluminium containers. The product should be stored between 5 °C and 30 °C.

Flammability

Not flammable under conditions of use. Not classified as a combustible liquid, as the boiling point (96°C) is less than the fire point (> 96°C). The product does not sustain combustion.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Exposure standards

NOHSC Exposure Standards:

Propylene glycol monomethyl ether (1-methoxy-2-propanol) TLV-TWA 100 ppm, 369 mg/m³, TLV-STEEL 150 ppm, 553 mg/m³

Exposure standard – Time Weighted Average (TWA) means the average airborne concentration of a particular substance when calculated over a normal eight-hour working day, for a five-day working week.

Exposure standard – Short term exposure limit (STEL) means a 15 minute TWA exposure which should not be exceeded at any time during the working day.

Biological limit

None allocated.

Engineering controls

Control process conditions to avoid contact. Use only in well-ventilated areas. If necessary, use local exhaust ventilation to keep airborne concentration below the exposure limits.

Personal Protective

Wear face shield or goggles

Equipment

Wear cotton overalls buttoned to the neck and wrist (or equivalent clothing) and a washable hat.

Wear elbow-length PVC or nitrile gloves.

If working in a poorly ventilated area or if occupational exposure levels are likely to be exceeded, wear a respirator suitable for organic vapours - AS/NZS 1715/1716 approved.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Blue to bluish-green liquid.

Odor: Weakly pungent.

Basic physical properties

Physical state: Liquid.

pH: 5.9 to 7.9

Density: 1.10 g/cm³ at 20°C

Melting point/freezing point: No data available.

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Flash point: Not detected till boiling at 101.6°C.

Evaporation rate: No data available.

Flammability (solid, gas): Not applicable.

Vapor pressure: <0.1mPa(20°C)

Solubility(ies): Soluble in water

Partition coefficient: n-octanol/water: Kow logP<0.1(pH 7, 22°C)

Auto-ignition temperature: No data available.

Decomposition temperature: No data available.

Viscosity: 0.24 Pa·s at 20°C; 0.10 Pa·s at 45°C.

Dynamic viscosity: No information available.

Explosive properties: No information available.

Oxidizing properties: No information available

10. STABILITY AND REACTIVITY

Chemical stability: Stable under normal conditions of use.

Conditions to avoid: Avoid sources of ignition and extreme heat.

Incompatible: Avoid contact with strong oxidising agents, acids or bases. Ammonia may be evolved in the materials presence of alkalis.

Hazardous: Ammonia. In a fire, oxides of carbon, nitrogen, phosphorus and sulphur may be emitted.

11. TOXICOLOGICAL INFORMATION

Acute studies:

Oral LD₅₀ (rat): >5000 mg/kg (slightly toxic).

Dermal LD₅₀ (rat): > 2000 mg/kg (slightly toxic).

Inhalation LC₅₀ (rat-4hrs): 3.73 mg/l (no more than slightly toxic)

Skin irritation(rat): slightly irritating.

Eye irritation(rabbit): moderately irritating.

Sensitization(Guinea pig): non-sensitizing.

The following data is based on the active ingredient.

Chronic toxicity: No clear evidence of target organ toxicity was noted in studies with rats, mice or dogs.

Carcinogenicity: None.

Teratogenicity: Glufosinate-ammonium, when administered to pregnant animals, produced no adverse effects on fetuses at doses which were not toxic to the dams.

Mutagenicity: No mutagenic activity was detected in a battery of mutagenicity tests.

Neurotoxicity: Does not inhibit acetylcholinesterase activities. No evidence of delayed neurotoxicity was noted in hens. Neurobehavioral effects (e.g., hypersensitivity, tremors, convulsions) related to stimulation of the central nervous system (CNS) were observed in some studies but only at lethal or near lethal dose levels.

Reproductive toxicity: Assessment of reproduction toxicity: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity: Assessment of repeated dose toxicity: The product has not been

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tested. The statement has been derived from the properties of the individual components.

12. ECOLOGICAL INFORMATION

Glufosinate-ammonium is practically non-toxic to birds, fish, honeybees, earthworms and beneficial insects. It is practically non-toxic to *Daphnia*, but moderately toxic to sensitive aquatic species. It is expected to be slightly to moderately toxic to wild animals. DO NOT contaminate streams, rivers or waterways.

Ecotoxicity	Glufosinate-ammonium:
<i>Fish toxicity:</i>	LC ₅₀ (96 h) rainbow trout 710 mg/L LC ₅₀ (96 h) carp, bluegill sunfish, golden orfe > 1000 mg/L
<i>Bird toxicity:</i>	Eight day dietary LC50 Japanese quail > 5000 mg/kg
<i>Daphnia toxicity:</i>	<i>Daphnia magna</i> EC50 (48 h) 560 – 1000 mg/L
<i>Algal toxicity:</i>	LD ₅₀ for <i>Scenedesmus subspicatus</i> ≥ 1000 mg/L LD ₅₀ for <i>Scenedesmus capricornutum</i> 37 mg/L
Environmental fate, persistence and degradability, mobility	Glufosinate-ammonium is very soluble in water and is hydrolytically and photolytically stable. It is rapidly degraded in surface levels of soils and in water. This product is considered to be readily biodegradable. The potential for groundwater contamination with glufosinate-ammonium is minimal. Glufosinate-ammonium does not accumulate in the fatty tissues of fish or other animals.

13. DISPOSAL CONSIDERATIONS

5, 10, 20 litre packs

Triple or preferably pressure rinse containers before disposal. Add rinsings to spray tank. Do not dispose of undiluted chemicals on site. If recycling, replace cap and return clean containers to recycler or designated collection point. If not recycling, break, crush or puncture and bury empty containers in a local authority landfill. If no landfill is available, bury the containers below 500 mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt. Do not re-use empty container for any other purpose. Dispose of waste product through a reputable waste contractor.

100, 110 litre packs

If tamper evident seals are broken prior to initial use then the integrity of the contents cannot be assured. Empty container by pumping through dry-break connection system. Do not attempt to breach the valve system or the filling point, or contaminate the container with water or other products. Ensure that the coupler, pump, meter and hoses are disconnected, triple rinsed and drained after each use. When empty, or contents no longer required, return the container to the point of purchase.

14. TRANSPORT INFORMATION

D.O.T:

UN number: N/A

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Shipping Description: Not Regulated by DOT

D.O.T. Hazard Class: N/A

Marine pollutant (Yes/No): No

IMDG/IMO:

UN number: N/A

UN proper shipping Name: Not Regulated

Hazard Class: N/A

Marine pollutant (Yes/No): No

ICAO/IATA:

UN number: N/A

UN proper shipping Name: Not Regulated

Hazard Class: N/A

Environm. Hazardous Mark: No

15. REGULATORY INFORMATION

U.S. FEDERAL REGULATORY INFORMATION

SARA TITLE III:

Section 302 (EHS) ingredients: Not listed.

Section 313 ingredients: Not listed.

SARA TITLE III NOTIFICATIONS AND INFORMATION

SARA TITLE III---HAZARD CLASSES: Acute Health Hazard---"Yes"

Chronic Health Hazard---"No"

Fire Hazard---"No"

Sudden Release of Pressure Hazard---"No"

Reactivity Hazard---"No"

EU REGULATORY INFORMATION

EU Classification: None

European Inventory of Existing Commercial Substances (EINECS)

Glufosinate-ammonium 77182-82-2

16. OTHER INFORMATION

This information is provided in good faith but without express or implied warranty. Buyer assumes all responsibility for safety and use not in accordance with label instruction.